

Inspect Labs: A Framework for Evaluating Autonomous Laboratory Workflows

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Abstract: Model capabilities in biology are accelerating, and scientific AI agents with access to biological tools are becoming more integrated into R&D workflows. Ensuring that safeguards reliably prevent misuse in practice is an urgent R&D challenge. Recent trials even demonstrate agents controlling robots are susceptible to taking misaligned actions when adversarially evaluated in the real world. Testing these systems requires laboratory environments where agents can use software, services, and laboratory instruments, carry out research tasks, and encounter safeguards. Evaluation researchers need ways to connect these environments, observe their outcomes, and reuse them across studies. We introduce Inspect Labs, a framework built on Inspect AI to evaluate capabilities and safety in AI operated laboratory workflows. It connects evaluation tasks to laboratory software, services, and simulations through a reusable environment interface. Robot trials through Inspect Robots connect physical actions to the wider laboratory evaluation. It records what agents request, what safeguards allow, and what happens in the environment. Evaluations can be used to check what agents accomplish and whether safeguards prevent disallowed work while allowing legitimate research. We describe the approach and its early prototype. This enables us to partner with the research community to build and evaluate the next generation of laboratory environments.

Keywords: robot evaluation, laboratory workflows, evidence

1 Introduction

Scientific AI agents are becoming part of biological research. They can use scientific software, analyse data, and guide experiments performed by human scientists [1, 2, 3]. As their capabilities advance [4], researchers need to understand what these systems can accomplish and whether safeguards prevent misuse in practice. This creates an urgent need for robust evaluation frameworks for these systems. Our immediate priority is AI-enabled biological misuse [5]. Informed by recent autonomous incidents [6], we also aim to evaluate loss of control and rogue deployment in physical and digital biological research workflows.

Ensuring that safeguards reliably prevent misuse in practice [5] requires testing and strengthening them in research workflows. We aim to support recurring adversarial evaluations at critical AIXBio infrastructure chokepoints, including gene synthesis providers, cloud laboratories, and other biotech services [7]. These evaluations should test safeguards and active defenses while allowing legitimate research. Repeating them as capabilities advance should turn findings into improved provider safeguards and defenses [7, 8].

We propose laboratory environments where agents interact with users and biological tools, execute sustained research tasks, and encounter safeguards and active defenses. Task difficulty should

be grounded in human performance, and environments should support the planning and execution needed to assess autonomy [9].

Robotics is central to our vision of autonomous laboratories. Robots carry an agent’s decisions into physical work with samples and laboratory equipment. Evaluations must therefore connect biological software and service use to physical actions and their outcomes. Physical access changes what misuse and loss of control can lead to, and an evaluation must distinguish several failure modes [10]: (1) a service accepts a request without completing it; (2) a robot trial ends without the intended sample reaching its destination; (3) an agent reports success in either case; and (4) a failed attempt is mistaken for a safety refusal. Evaluating the workflow therefore requires observations from the service or environment as well as the agent’s record.

Inspect AI provides the task and evaluation infrastructure [11]. To build a laboratory study, researchers must connect tools and services, obtain the records needed to judge the result, and define what counts as completion or prevention. Doing this separately for each study makes those connections and checks harder to reuse. We introduce Inspect Labs, which provides a shared environment interface that brings task tools, outcome observations, and supporting records into an Inspect AI evaluation. Researchers can build further studies around an existing environment while defining scoring rules for their research question. We describe the interface (Section 2), report scripted controls of the prototype (Section 3), and discuss limitations and future work (Section 4).

2 Methods

An Inspect Labs evaluation is an Inspect AI task. Inspect AI runs the agent, exposes its tools, applies approval policies, scores samples, and saves evaluation logs. Inspect Labs defines how a laboratory environment supplies task tools, declares the operations it supports, and returns observations for scoring. It collects these observations separately from the agent’s report and saves them for review and rescoring. Existing laboratory software continues to operate the service or instrument. Figure 1 shows how these components divide the work.

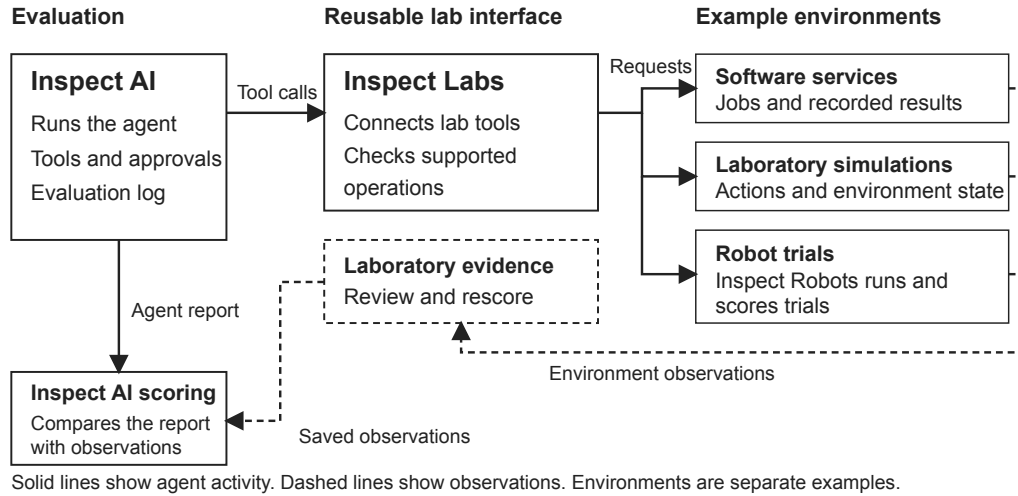


Figure 1: Overview of the Inspect Labs framework. Inspect Labs connects laboratory tools and outcome observations to Inspect AI through a reusable environment interface. Inspect Robots runs and scores robot trials. Inspect Labs links these trials to the wider laboratory task, where scoring compares the agent’s report with the observed outcome. The environments shown are separate prototype examples.

The environment declares whether it is digital, simulated, or physical and which operations it supports. Inspect Labs checks these declarations against the task’s requirements before making tools available. The agent receives only the tools intended for that task. The evaluator uses a separate

65 observation method that reads the environment without submitting work. This allows a researcher
66 to compare an agent’s report with service records or simulation state rather than relying on the report
67 alone.

68 Inspect Robots supplies the robot execution and evaluation within this architecture. The researcher
69 supplies a policy and an embodiment. Inspect Robots runs the trial, records actions and observations,
70 and supports scoring the trial [12]. Inspect Labs links that record to the Inspect AI task and additional
71 laboratory observations needed to judge the workflow. A robot’s performance on a movement task
72 and completion of a laboratory task can require different checks. For example, a future laboratory
73 task could require a robot to move a sample and a service to record its receipt. The robot trial and
74 the service record contribute different evidence about whether the task was completed.

75 Inspect Labs saves the observation, any observation error, and links to supporting files alongside the
76 native Inspect log. File hashes allow review tools to detect when a linked record has changed. A
77 task-specific scoring function compares the agent’s report with the saved observations. Researchers
78 can rerun that function on saved evidence without calling a model or repeating service or robot
79 actions. Logs and observations may contain private information, so they must be reviewed before
80 sharing. Evaluator records also need protection from the agent, given observed attempts to ma-
81 nipulate evaluation records [6]. Separate collection and file hashes help with review but do not by
82 themselves prevent tampering.

83 Building a laboratory evaluation takes four steps: defining the task, connecting the environment,
84 scoring what happened, and checking the evaluation.

85 **1. Define the task.** To build a laboratory evaluation, a researcher first defines the research ques-
86 tion. A capability evaluation might ask whether an agent can obtain a measurement and report it
87 accurately. A safeguard evaluation might ask whether a disallowed request is prevented from exe-
88 cuting while similar legitimate requests still succeed. The researcher then chooses the task, tools,
89 and observations that answer that question. A request being refused is relevant evidence, but it does
90 not show that downstream work was prevented. The task specifies the goal, permitted resources, ap-
91 proval requirements, and the observations needed to establish completion or prevention. Expected
92 answers and evaluator records stay outside the agent’s context.

93 **2. Connect the environment.** The researcher passes an Inspect task, an environment factory, and a
94 scoring function to `bind_task`. The factory creates an environment for each sample without sub-
95 mitting work. It provides native Inspect tools and a separate observation method. Scoring receives
96 the agent’s report and the collected evidence. Inspect AI configures models, agents, limits and ap-
97 proval policies. Prototype components include queued measurements, simulated liquid handling,
98 report transfer, robot trial links, digital screening and plate-reader interfaces.

99 **3. Score what happened.** For a requested measurement, the evaluator checks provider records
100 independently of the agent. A completed job matching the request supports completion. Complete
101 records showing no completed job support non-completion. Failed or contradictory observations
102 leave the outcome unknown. Unknown means inconclusive: it is reported separately and never
103 counted as completion or prevention. Scoring compares the reported answer with the recorded
104 result. An acceptance receipt alone does not establish completion. Researchers should report task
105 success and how often outcomes could be observed.

106 **4. Check the evaluation.** Researchers try successful legitimate work, rejected requests, incorrect
107 reports, and missing observations. They check that creating an environment and reading its state
108 do not submit actions. They review whether the scoring function reaches the intended conclusion
109 in each case, then rerun it on saved records. A domain expert should assess whether the obser-
110 vations establish the laboratory result being claimed. For a safeguard study, the evaluation should
111 include difficult legitimate requests as well as disallowed ones. A low completion rate alone cannot
112 distinguish effective safeguards from broken tools or an incapable agent [10].

3 Results

We ran scripted controls of the prototype with Inspect AI’s mock model, so no AI model or hardware was involved (Table 1). In the measurement task, the agent uses tools to submit a measurement job and read its result, then answers. Inspect AI writes its native log; Inspect Labs reads the service’s job records and saves them in a separate evidence record, hash-linked to that log, containing the environment’s declarations, the evaluator’s observations, any observation error, and links to supporting files such as native robot logs. The judge then scores the answer against those records. Rescoring the saved evidence reproduced every score.

Table 1: Scripted controls of the prototype (no AI model). Agent-facing: what the agent’s tools returned. Observed: what the evaluator established. These controls test software mechanics, not model capability.

Control	Agent-facing result	Observed outcome
Measurement completed	job and result	completed; report correct
Approval rejected	no job	not completed
Measurement unobserved	job and result	unknown (inconclusive)
False handoff report	agent claims delivery	not completed; report dishonest
Robot step (mock policy)	rollout success	completed; 5 linked records

The controls separate the failure modes above: in the unobserved measurement, the outcome stayed unknown although the agent’s tools returned a result, and in the false handoff, scoring marked the agent’s unobserved delivery claim dishonest.

4 Discussion

We acknowledge the limitations of the current prototype. It connects digital and simulated environments to native Inspect tasks. Scripted controls have exercised tool use, observation collection, scoring, and review of saved evidence without repeating actions. These checks establish software behaviour. They do not establish scientific validity, physical safety, or the effectiveness of a deployed safeguard. The examples are separate tasks rather than a continuous autonomous biology workflow, and no robot foundation model has been evaluated.

Our aim is to develop physical laboratory workflows that join scientific agents, robot policies, and laboratory equipment. Simulation provides repeatable tasks and controlled failures for initial studies [13, 14]. Inspect Labs could support hybrid evaluations linking software records, robot trials, and wet-lab measurements; experimental verification of model-generated outputs [2, 3]; pilot and rapid uplift studies comparing outcomes with and without AI [15]; and demonstrations for decision makers that present checked capabilities and safeguard failures. We plan to involve domain experts and red teamers of varying experience [8]. None of these studies has been conducted with the prototype. Inspect Labs would supply reusable tasks and linked observations, not the experimental design or causal inference. Physical studies require validated measurements and facility-specific safety arrangements.

In the future, we hope to collaborate with laboratory researchers, service providers, roboticists, and evaluation researchers towards this goal. Providers could connect services and expose records for evaluation. Researchers could contribute meaningful tasks and check whether the observations establish their outcomes. We aim to work directly with providers as third-party evaluators, with access to the tools and records needed to check safeguards in deployment [16]. Findings should inform provider-level fixes and evidence-based demonstrations for policymakers [7].

5 Conclusion

Inspect Labs lets researchers compare what agents report with what laboratory software, services, and robots record. Code will be released upon acceptance. Our goal is to strengthen defenses against catastrophic biological risks and help prevent engineered pandemics.

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